

Research Question

What factors among **Biological Sex, School Safety, and Risk Alcohol Use** are significantly associated with the proportion of high school students who felt sad or hopeless?

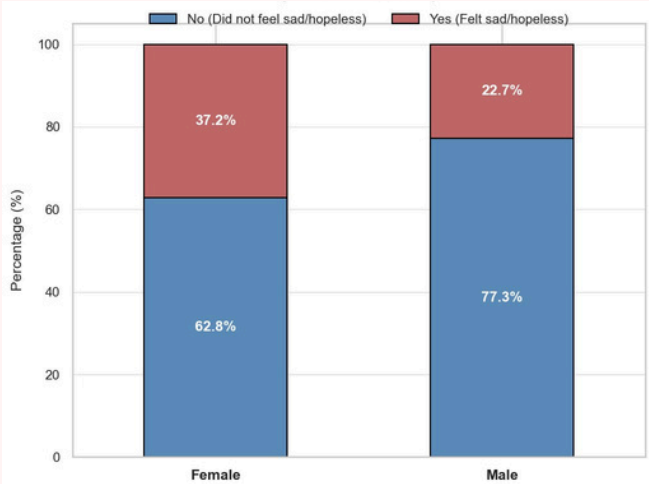
Method

- Statistical Method: Independent Two-Proportion Z-test & 95% Confidence Intervals
- Significance Level: 0.05
- Statistical Hypotheses :
H_0: $p_1 - p_2 = 0$
(The specific factor has no significant association with the proportion of student sadness).
H_a: $p_1 - p_2 \neq 0$
(The proportion of students who felt sad or hopeless differs significantly between the two groups).

Statistical Data

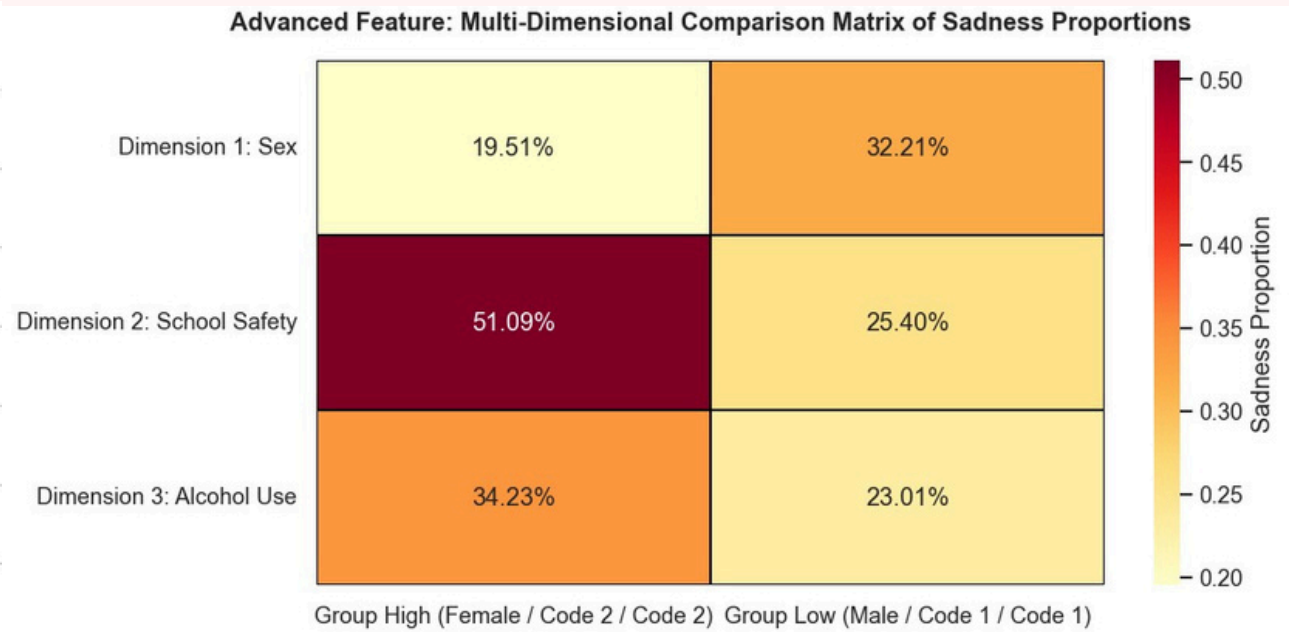
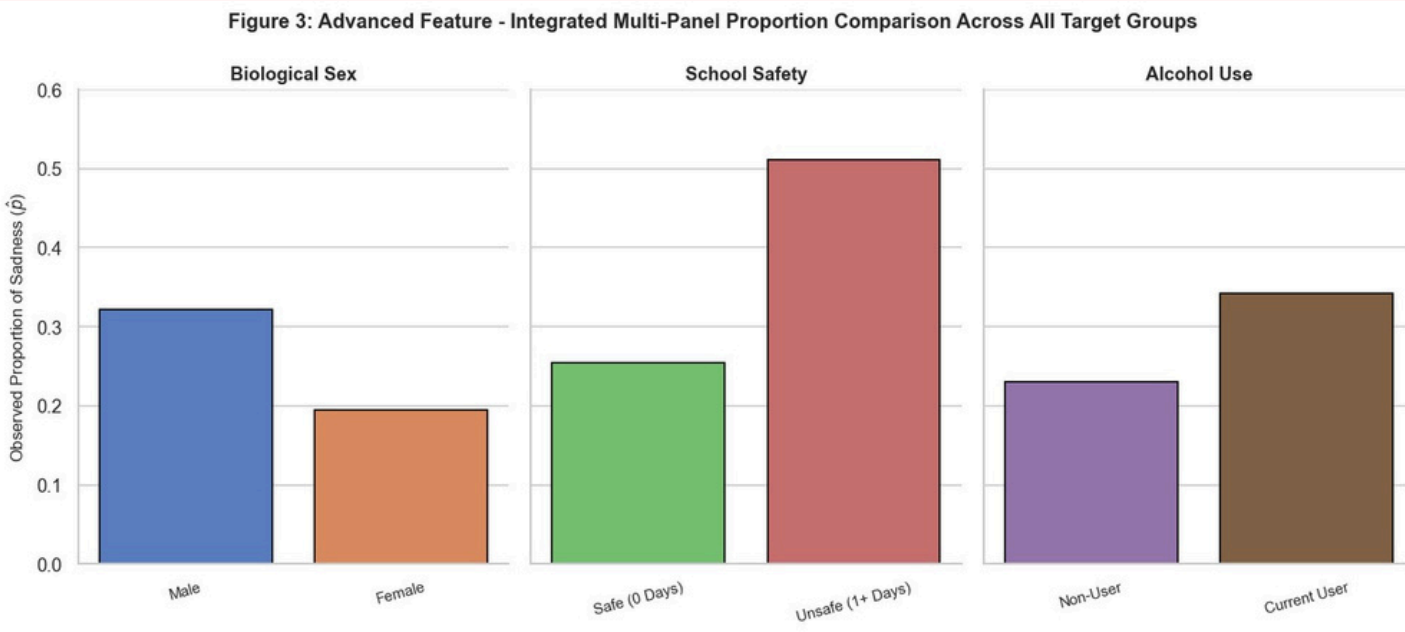
Research Dimension	Difference (p1-p2)	Z Statistic	P-Value
D1: Biological Sex (Female vs Male)	-0.1270	-13.9821	2.00E-44
D2: School Safety (Code 2 vs Code 1)	0.2569	9.5657	1.11E-21
D3: Alcohol Use (Code 2 vs Code 1)	0.1123	11.1147	1.06E-28

Results



The data shows a significant gender gap, with the proportion of adolescent females experiencing emotional distress being considerably higher than that of males (37.2% vs. 22.7%).

Key Results



1. School Safety : Sadness proportion is 51.09% for the unsafe group vs. 25.40% for the safe group (25.69% gap)
2. Biological Sex : Sadness proportion is 32.21% for females vs. 19.51% for males (12.70% gap).
3. Alcohol Use : Sadness proportion is 34.23% for users vs. 23.01% for non-users (11.23% gap).

CONCLUSION

- Environmental safety is the strongest predictor of adolescent mental health crises.
- Females exhibit a long-term, stable baseline of higher risk during adolescence.
- Alcohol use and emotional distress frequently co-occur.